The Alpha Difficulty Scoring Model

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Abstract

Alpha is a first, intentionally simple model for scoring the navigational difficulty of a route through an indoor orienteering venue. It is additive over two structural quantities — stair crossings and branching points — and its scale is calibrated so that scores align with the difficulty categories published by Stockholm Indoor Cup [2]. This note defines the model in isolation, as a baseline against which future scoring models can be compared; it is a condensed excerpt of the difficulty-scoring model introduced in the companion article on graph-based indoor orienteering analysis [1].

1 Model definition

Alpha assumes a route has already been identified (as a sequence of graph edges) and scores it from two counts:

$$D \;=\; k\big(5\,N_f + 3\,N_b\big), \qquad k = \frac{100}{81}$$

where

- N_f is the number of stair (floor-change) traversals along the route, upward and downward combined — an edge counts as a stair when its endpoints resolve to different floor levels;
- N_b is the number of **branching points**: route nodes at which more than one next node is reachable, excluding the node just come from and respecting one-way constraints;
- $k = 100/81$ is a scaling factor that normalizes a reference leg (Stockholm Indoor Cup 2023, stage 1, course 1, leg 3→4; raw score 81) to exactly 100.

Leg difficulty is the score of the leg's shortest route; course difficulty defaults to the *maximum* leg difficulty. Scores map onto the five-color scale at fixed thresholds (Table 1). Alpha is implemented as part of the open-source analysis toolchain [3], which computes it directly from a venue graph.

Category	Score range
White	0–20
Yellow	20–40
Orange	40–60
Purple	60–80
Black	80+

Table 1: Difficulty scores mapped to difficulty categories.

2 Worked example

Figure 1 shows a two-floor fragment with a single route from start S to finish F. The route crosses two labeled stairs (A and B) and passes through one branching point, where a dead-end corridor is also reachable.

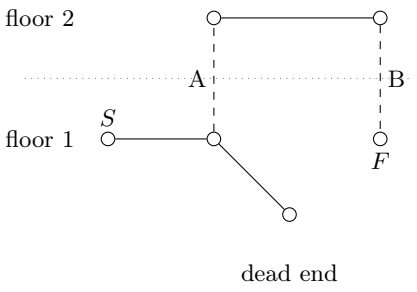


Figure 1: Worked example. The route from S to F crosses stairs A and B ($N_f = 2$) via one branching point — the node where the dead-end corridor also departs ($N_b = 1$). This gives $D = k(5 \cdot 2 + 3 \cdot 1) = \frac{100}{81} \cdot 13 \approx 16.0$, in the White category.

3 Refinements beyond raw branching

Raw branching counts every fork, including trivially dismissible ones (such as the dead end in Figure 1, which is short and unlikely to mislead). Two refinements are computed alongside the score and available to future models:

- **Dead-end filtering:** edges on non-branching chains that terminate at a leaf are excluded — a passage with only one way in and out creates no real decision.
- **Decision points:** a stricter variant that counts a node only when an off-route branch is *significant*, i.e. runs at least 50 m before dead-ending or rejoining the route. A 10 m alcove is visibly wrong and creates no decision; a 50 m parallel corridor is a genuine route choice.

4 Known limitations

Alpha is a baseline, and its limitations chart the research agenda: branching is counted, not weighted by how hard the alternatives are to reject; the additive form cannot express the combinatorial explosion of several *sequential* decisions; and the score depends on how densely the navigation graph is drawn, so graph-drawing conventions must be standardized before scores are comparable across venues. These are taken up as research directions in the companion article [1].

References

[1] *A graph-based framework for quantitative analysis of navigational complexity in indoor orienteering courses* (companion article).

[2] Stockholm Indoor Cup. *Svårighetsgrader — Difficulty levels.*

[3] *Indoor Orienteering Route Analysis* (source code, algorithm reference, and datasets). `github:objorke/indoor-o-route-analysis`.